

Welder

- Fundamentals of welding processes and safe workplace organisation.
- Joint preparation, equipment, protection and inspection.
- A training programme for work in the Republic of Kazakhstan.

This course does not automatically award a vocational grade.



Course content

- The course covers arc welding, gas flame processes and equipment.
- Separate sections address electrical, gas and fire hazards.
- Practical scenarios connect preparation, execution and completion.

The welding procedure specifies parameters for the actual task.

| Purpose of a welded joint

- Welding creates a permanent joint between materials.
- Required properties depend on the material, structure and operating conditions.
- Weld appearance does not verify all internal properties.
- Acceptance follows the requirements for the specific product.

**Structural safety
depends on joint
quality.**



| Main processes

- An electric arc provides heat in arc welding.
- Gas welding uses the flame of fuel gas and oxygen.
- Weld-pool protection depends on the selected process.
- Select the process according to material, thickness and procedure.

One set of parameters cannot apply to every process.

| Manual metal arc welding

- The consumable electrode contributes to the weld metal.
- Its coating provides the intended process protection.
- Select the electrode classification, diameter and parameters using the welding procedure.
- Electrode condition affects joint quality.

Damp or unsuitable materials must be replaced.



| Gas-shielded welding

- The process uses wire or a non-consumable electrode, depending on its type.
- Gas shields the weld pool from the surrounding atmosphere.
- Interrupted gas supply can impair quality.
- Select the gas and flow rate according to the procedure and equipment.

Shielding gas can displace oxygen from the work area.

| Gas welding and cutting

- The torch or cutting torch must match the gas used.
- Gas welding and oxygen cutting perform different tasks.
- Select parameters and equipment according to the material and manufacturer's instructions.
- Use the lighting and shutdown sequence for the actual equipment.

**There is no
universal pressure
setting for all
torches.**



Task documentation

- Read the drawing and joint requirements before work.
- Establish the material, edge preparation and welding position.
- Check the procedure, parameters and inspection requirements.
- Clarify unclear requirements before welding starts.

An informal assumption cannot replace the specified procedure.



| Preparing components

- Clean surfaces as required by the procedure.
- Remove contamination that could impair quality or produce harmful emissions.
- Check edge shape and joint gap.
- Secure components for stable positioning and safe access.

Assess an unknown coating before heating it.



| Assembly and tack welds

- Verify component alignment against the drawing.
- Make tack welds according to material and procedure requirements.
- Fixtures must hold components without dangerous movement.
- Remove unacceptable tack-weld defects using the established method.

**Tack welds also
affect joint quality.**

| Parameters and sequence

- Select current, voltage and travel speed using the procedure.
- Account for welding position, thickness and the component's thermal condition.
- Any parameter change must be justified.
- Pass sequence helps control shape and distortion.

**Guessing
parameters is
especially
dangerous for
safety-critical
structures.**

| Welding power sources

- Transformers, rectifiers and inverters have different characteristics.
- Verify power supply and permitted load against the equipment passport.
- Follow the cooling and duty-cycle instructions.
- Stop in the event of overheating or a fault.

**A compact machine
still requires correct
connection.**



Cables and electrode holder

- Check insulation, connections and electrode clamping.
- Protect cables from heat, cuts and moving vehicles.
- Place the holder to prevent accidental short circuits.
- Remove defective components from use.

Correct damaged insulation before energising.

| Work return conductor

- Current must return through the intended welding circuit.
- Provide a reliable connection to the workpiece.
- Do not substitute random pipes or structures for the return conductor.
- Apply the special insulation requirements in fire or explosion hazardous conditions.

The welding circuit and protective earthing serve different purposes.

| Permanent and temporary welding stations

- A permanent station is equipped for repeated operations.
- A temporary station requires assessment of the actual task location.
- Access, lighting and equipment positioning must be safe.
- Coordinate nearby activities before starting.

**Moving the machine
creates a new
workplace.**



| Main welding hazards

- The arc creates radiation and electrical shock hazards.
- Sparks and hot metal cause burns and fires.
- Fumes and gases affect breathing.
- Handling components, noise and awkward posture create additional risks.

**Select protection for
all hazards of the
task.**

| Eye and face protection

- Select a welding helmet or shield for the process and parameters.
- The filter must provide the required protection.
- Check integrity, attachments and function.
- Use protection against flying particles during grinding and cleaning.

**Ordinary glasses
cannot replace a
welding filter.**

| Clothing and hand protection

- Clothing must protect against sparks and hot particles.
- Use welding gloves and keep them dry.
- Prevent exposed skin and pockets that collect sparks.
- Footwear and clothing arrangement must keep hot metal out.

**Clothing
contaminated with
combustible
substances is
unsuitable.**



| Protecting other people

- Separate the welding area with the specified screens.
- Prevent unprotected people from looking directly at the arc.
- Check where sparks and particles travel.
- Keep access routes clear and easy to identify.

**Arc hazards extend
beyond the welding
station.**



Fumes and ventilation

- Remove contaminants as close to their source as possible.
- General ventilation supplements local extraction.
- Emissions depend on the metal, coatings and consumables.
- Select respiratory protection from a risk assessment; it does not replace required ventilation.

The absence of a strong smell does not prove clean air.

| Electrical safety for welders

- Authorised personnel connect the machine using the established method.
- Check protective connections and cables before work.
- Do not work with damaged equipment or unsafe moisture conditions.
- Repairs and switching require the relevant separate authorisation.

An electrical safety group does not authorise every operation.



| Pre-shift checks

- Clarify the task, documentation and authorisation.
- Inspect the machine, cables, fixtures and protective equipment.
- Check ventilation, screens and fire precautions.
- Correct deficiencies before welding.

The check applies to today's location and task.



| Preparing hot work

- Obtain the prescribed authorisation for temporary hot work.
- Remove or protect combustible materials.
- Check openings and adjoining spaces.
- Provide firefighting equipment and the required monitoring.

**Sparks can cause
smouldering out of
sight.**



Oxygen and fuel gases

- Oxygen supports combustion and increases fire hazards.
- Fuel gas can form a dangerous mixture with air.
- Keep oxygen fittings free from oil and grease.
- Never use oxygen to blow off clothing or ventilate a space.

Each gas requires compatible equipment.

| Identifying a cylinder

- Identify the gas from markings and documentation.
- Check the shell, valve and required examination information.
- Do not use a defective or uncertainly identified cylinder.
- Colour alone is not sufficient for connection.

**Uncertainty about
the contents rules
out connection.**



| Positioning and moving cylinders

- Secure cylinders against falling and protect them from heat.
- Use equipment intended for cylinder handling.
- Do not lift by the valve or expose cylinders to impacts.
- Determine separation distances and co-location from current requirements.

Valve damage can cause a sudden release.



| Regulators and gauges

- Select a regulator for the gas and pressure range.
- Use compatible connections without improvised adapters.
- Check gauge condition and absence of damage.
- Set operating pressure according to the torch instructions.

**A value from
someone else's
example is not a
setting.**



Hoses and connections

- Use hoses suitable for the gas and working conditions.
- Protect them from kinks, sparks and mechanical damage.
- Use the specified fasteners for connections.
- Check leaks by an approved method without an open flame.

**A smell or
suspected leak
requires stopping.**



| Flashback and protection

- Flashback carries the flame back towards the gas system.
- Specified valves and flashback arrestors reduce the risk.
- Stop according to the instructions after a pop, overheating or unstable operation.
- Restart only after establishing and correcting the cause.

A safety device does not replace a serviceable system.

| Welding containers

- Before heating, establish what the container previously held.
- Remove residues and isolate connected services under the procedure.
- Test the atmosphere and complete the required authorisation measures.
- Do not independently weld a closed or unidentified container.

An apparently empty container may contain flammable vapours.

| Confined spaces

- Entry requires special preparation and established authorisation.
- Arrange atmosphere testing, ventilation, communication and an attendant.
- Plan rescue without improvisation.
- Stop and leave if conditions change.

Shielding gas can create an oxygen-deficient atmosphere.



| Work at height

- Use a safe platform and the specified fall protection.
- Select attachment points under the established system, not arbitrarily.
- Protect fall-protection components from sparks and hot metal.
- Barrier off the lower area against falling objects and particles.

A random scaffold component is not a verified anchorage.

| Handling components

- Determine mass and the handling method beforehand.
- Use suitable lifting equipment and stable supports.
- Keep hands clear of potential trapping points.
- Identify hot components and place them safely.

**Metal appearance
does not always
reveal its
temperature.**

| Making the weld

- Maintain the parameters and sequence specified by the procedure.
- Monitor process stability and equipment condition.
- Carry out the required cleaning between passes.
- Do not continue if equipment fails or protection is lost.

Stop for unsafe conditions before a defective joint results.

| Visible joint defects

- Cracks, undercut, overlap and unfilled craters require assessment.
- Weld size and shape must match the documentation.
- Porosity and lack of penetration can have different causes.
- Determine acceptance from requirements, not personal impressions.

**Grinding does not
prove removal of an
internal defect.**



| Inspection and repair

- Supplement visual inspection with the required methods.
- Record results against the specific joint.
- Use an agreed procedure for repair.
- Repeat inspection to confirm the repair result.

**Unauthorised repair
can conceal the
cause of a defect.**



Breaks and completion

- Shut down equipment and the gas system as instructed.
- Prevent unintended starting and leaks.
- Inspect the work area and adjacent spaces for smouldering.
- Record completion and maintain the required fire monitoring.

Cooling and concealed smouldering require attention after welding.

| Faults or leakage

- Stop work and carry out a safe shutdown.
- Warn people and notify the responsible person.
- Do not search for gas leaks with a flame or repair under pressure.
- Resume only after the cause has been confirmed as corrected.

Repeated popping is not normal operation.



Fire and emergency notification

- Report the exact location and nature of the hazard.
- Call for help under the established procedure; use 112 or 101 where necessary.
- Maintain a safe escape route.
- Attempt firefighting only with suitable training and safe conditions.

Evacuate when fire threatens to spread.

| First aid and safety

- First stop harmful exposure without endangering yourself.
- Call emergency services and act within your training.
- For electrical injury, do not touch the person until electrical exposure is safely removed.
- Do not enter a hazardous atmosphere without protection to attempt rescue.

**Assistance requires
safe conditions.**

| Scenario: an unidentified container

- You are asked to weld an attachment onto a used container.
- Its previous contents are unknown.
- Stop until identification, preparation and authorisation are complete.
- External cleanliness does not prove absence of hazardous residues.

Heat only after safe conditions are verified.



Final knowledge assessment

- The welding procedure determines materials, preparation and parameters.
- Check equipment and protective measures before work.
- Faults, leaks and changed conditions require stopping.
- Select Start test.

**Safe completion is
part of every
welding task.**